House Price Prediction – Project Summary

Overview

The objective of this project was to develop a machine learning model capable of predicting house prices using various property features. The dataset used for this project was obtained from Kaggle and contained information such as area, number of bedrooms, bathrooms, parking availability, stories, furnishing status, and other amenities. The target variable for prediction was the house price.

Data Preparation

The dataset was first explored to understand its structure and identify any missing values or duplicate records. Duplicate rows were removed and categorical variables such as main road access, guest room availability, basement, air conditioning, and furnishing status were converted into numerical format using one-hot encoding. After preprocessing, the dataset was ready for machine learning model training.

Model Development

The data was divided into training and testing sets using an 80:20 split. Two regression models were trained and evaluated:

1. Linear Regression
2. Random Forest Regressor

The performance of both models was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Model Performance

Linear Regression

- MAE: 970,043
- RMSE: 1,324,507
- R^2 Score: 0.653

Random Forest Regressor

- MAE: 1,013,969
- RMSE: 1,398,116
- R^2 Score: 0.613

Based on these metrics, the Linear Regression model performed slightly better than the Random Forest model and was selected as the better-performing model for this dataset.

Key Findings

The analysis showed that features such as house area, number of bathrooms, parking availability, number of stories, and air conditioning had a strong influence on house prices. The correlation analysis indicated that larger houses generally have higher prices, while additional amenities also contribute significantly to property value.

One interesting observation was that the Random Forest model did not outperform the Linear Regression model, which suggests that the relationship between the available features and house prices is relatively linear. The distribution of house prices was also right-skewed, indicating the presence of a small number of high-value properties.

Conclusion

This project successfully demonstrated how machine learning can be used to estimate house prices based on property characteristics. The Linear Regression model achieved an R^2 score of approximately 65%, indicating that it was able to explain a significant portion of the variation in house prices. Such predictive models can assist real estate businesses in pricing properties more accurately and making data-driven decisions. Future improvements could include collecting additional location-specific features and testing more advanced regression techniques to improve prediction accuracy.